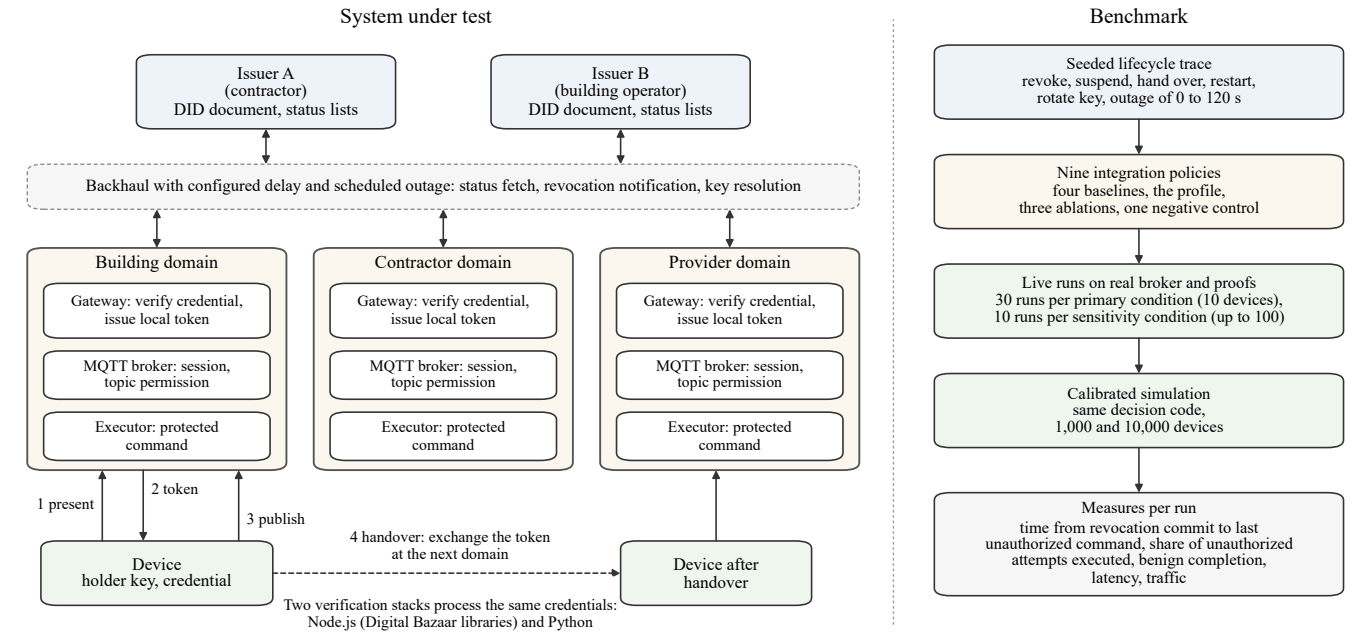
System under test
Issuer A
(contractor)
DID document, status lists
Issuer B
(building operator)
DID document, status lists
Backhaul with configured delay and scheduled outage: status fetch, revocation notification, key resolution
Building domain
Gateway: verify credential, issue local token
MQTT broker: session, topic permission
Executor: protected command
Contractor domain
Gateway: verify credential, issue local token
MQTT broker: session, topic permission
Executor: protected command
Provider domain
Gateway: verify credential, issue local token
MQTT broker: session, topic permission
Executor: protected command
1 present
2 token
3 publish
Device
holder key, credential
4 handover: exchange the token at the next domain
Device after handover
Two verification stacks process the same credentials:
Node.js (Digital Bazaar libraries) and Python
Benchmark
Seeded lifecycle trace
revoke, suspend, hand over, restart, rotate key, outage of 0 to 120 s
Nine integration policies
four baselines, the profile, three ablations, one negative control
Live runs on real broker and proofs
30 runs per primary condition (10 devices), 10 runs per sensitivity condition (up to 100)
Calibrated simulation
same decision code, 1,000 and 10,000 devices
Measures per run
time from revocation commit to last unauthorized command, share of unauthorized attempts executed, benign completion, latency, traffic